6
The Evolution of Materials and Architecture for Oxygen Transport Membranes

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6.1
Introduction

In this chapter we discuss progress made in ceramic oxygen-ion transport materials, their use in devices for separating oxygen from air with the purpose of collecting oxygen, and the use of oxygen separation membranes for partial or complete oxidation of fuel species. The requirements for materials used for oxygen separation and collection have similarities with, but also some differences from, materials used in partial oxidation (for the production of syngas) or complete oxidation (combustion). Both sets of membrane materials require substantial ionic and electronic conductivity as well as high-temperature stability and mechanical strength. However, the materials used for production of syngas and in combustion applications require a tolerance to high oxygen partial pressure on the airside, low oxygen partial pressure on the fuel side, and chemical resistance to steam, CO_2, H_2 and CO. The differences in requirements have led to the development of subsets of mixed electronic-ionic conductors (MEIC) suitable for oxygen separation or syngas production.

In parallel to development activities in the materials themselves, significant progress has been made in the architecture of devices. This includes a transition from dense, self-supported membranes to the use of thin films supported on porous substrates, as well as developments in tubular geometries and modified planar geometries. This chapter will provide an outline of these development activities and provide possible insight into potential future developments.

6.2
Oxygen Separation and Collection

We will first consider the development activity which has occurred in the application of oxygen separation and collection. A wide range of materials has been examined in this area including perovskites, brownmillerites, fluorites and composites of various structures. There has also been progress in membrane architec-

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ture to provide high oxygen fluxes with the structural strength necessary for commercial applications.

6.2.1 Background for Selection of Materials for Oxygen Separation and Collection

It is important to understand the forces driving materials development activities within this field. Areas of development have included attempts to increase oxygen-ion conductivity, to increase surface exchange rates, to decrease creep rates, and to increase tolerance to CO_2.

The oxygen transport membrane (OTM), alternatively called an ion transport membrane (ITM), is a key component in an oxygen separation device. The membrane materials allow the diffusion of oxygen anions and potentially produce infinite selectivity for oxygen obtained from gas mixtures containing oxygen, such as air. Oxygen diffusion is driven by an oxygen partial pressure gradient across the membrane. Commonly, in laboratory experiments, oxygen gradients are maintained by using a sweep gas such as helium in the permeate. In practical oxygen separation devices, the oxygen gradient is maintained by one of the following procedures: (i) pressurizing air in the feed stream, (ii) producing a vacuum on the membrane permeate side, (iii) using a sweep gas such as steam which can be separated readily from the oxygen product through condensation.

Each process has its advantages and disadvantages. Pressurizing the air feed stream allows the collection of oxygen in the permeate at or above ambient pressure with no need for further processing. However, the feed air stream typically needs to be highly pressurized (e.g., 20 bar (280 psi) absolute), leading to substantial stresses on the membrane as well as to high energy and capital costs associated with compressing large volumes of air. Using a vacuum on the permeate side produces only a relatively low absolute pressure differential across the membrane. However, typically, the product gas must be re-compressed to a useful working pressure because piping of a low-pressure gas can become uneconomical for industrial and power plant applications. The option of using steam as a sweep gas has the advantage of reducing the absolute pressure differential requirements on the membrane, without requiring excessive compression of the oxygen. However, such membrane materials must be tolerant to a high partial pressure of steam. The majority of effort made in membrane materials development has been directed at the high-pressure feed option.

A typical oxygen separation device may operate with an absolute feed air pressure of 20 bar and collect oxygen at approximately 1 bar (14.5 psi). Thus, the inlet partial pressure of oxygen on the membrane feed side is approximately 4 bar (58.0 psi). If 50% of the oxygen in the feed is to be utilized, the outlet of the feed stream would have an oxygen partial pressure of 2 bar (29.0 psi). Thus, in rough terms, the average oxygen partial pressure ratio of the feed stream to the product stream is 3:1. In a solid oxide fuel cell, the oxygen partial pressure ratio is approximately 10^{15}:1. It is clear that the oxygen gradient, and thus the driving force, is much lower in oxygen transport membranes than in solid oxide fuel cells. As a

result, there has been a dedicated effort to improve the other properties which increase the oxygen flux, namely ionic conductivity and surface exchange rates.

Most oxygen-ion transport membranes operate in the temperature range 750 to 1000 °C. This creates several problems for operation of ceramic membranes. First, the membranes must tolerate a 20 bar pressure differential at temperatures that are conducive to generating creep-induced failure. The membrane materials which have high ionic conductivity also have large numbers of lattice vacancies to facilitate mobility of the ionic species, but lattice vacancies are major contributors to elevated creep rates.

There are several options to heat the inlet gas. These include directly heating the gas by the injection and oxidation of a fuel such as natural gas, or indirectly heating the gas by means of heat exchange. There are merits and disadvantages to both. However, from a materials selection perspective, directly heating the gas by oxidation results in a feed stream which contains significant amounts of carbon dioxide and steam. Thus, when selecting the membrane material, the heating method must be carefully considered.

The ceramic membrane typically must also be joined to a metal manifold to collect the product gas. The requirements of ceramic–metal joints are extremely stringent due to the large pressure differential across the membrane. Even very small leaks will create a drop in product purity. The joints are usually at the operating temperature, although cold seals are occasionally considered. The thermal expansion coefficient of the ceramic needs to match that of the metal manifold. Table 6.1 summarizes the materials properties desired for operation of a pressure-driven ceramic oxygen transport membrane.

Tab. 6.1 Desired materials properties for ceramic oxygen transport membranes used for oxygen separation

Property	Maximum	Minimum	Preferred range
Oxygen ion conductivity (S cm^{-1})	N/A	1	2–4
Electronic conductivity (S cm^{-1})	N/A	5	20–500
Oxygen surface exchange coefficient (cm s^{-1})	N/A	10^{-7}	10^{-5}–10^{-7}
Thermal expansion coefficient (ppm K^{-1})	20	8	12–17
Chemical expansion coefficient (%)	0.25	–0.25	0–0.1
Creep rate (cm s^{-1})	10^{-8}	N/A	10^{-9}–10^{-11}
CO_2 tolerance (bar CO_2)	N/A	0.1	0.6–1.0

6.2.2
Membrane Materials Concepts

There are two main approaches for fabricating oxygen separation membranes. The first is to use a mixture of two distinct materials – an oxygen-ion conductor such as doped zirconia, and an electron conductor such as a noble metal or a doped perovskite (see Fig. 6.1). The second approach is to use a material which is capable of conducting both oxygen ions and electrons (see Fig. 6.2).

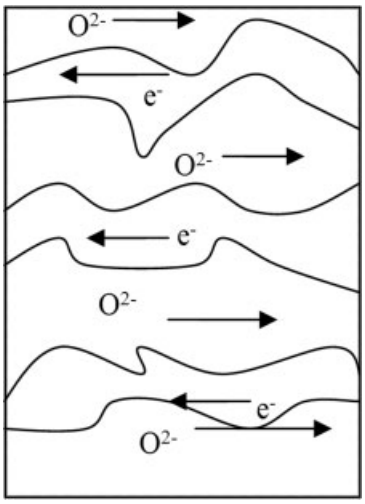


Figure 6.1 Dual-phase membrane.

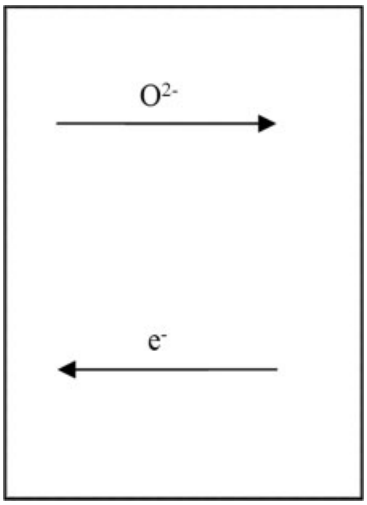


Figure 6.2 Mixed electron-oxygen ion conducting membrane.

The first approach is known as the dual-phase membrane. Much of the materials selection can be directly taken from solid oxide fuel cell work. The most commonly studied materials for the ion conducting electrolyte are YSZ (yttria stabilized zirconia) [1, 2], CGO (cerium gadolinium oxide) [2] and Bi_2O_3 [3]. The electron-conducting materials have generally been noble metals due to problems with chemical interaction, particularly in the cases of Bi_2O_3 and YSZ. However, there is also work to incorporate electron-conducting perovskite ceramics such as those based on $LaCoO_3$ [4].

There are several advantages to using a dual-phase membrane over a single-phase mixed electronic ionic conductor. These include the fact that the ionic conductors such as YSZ are much more chemically and thermally stable compared to most perovskites. Thus, dual-phase membranes are likely to be able to tolerate the harsh conditions of an oxygen separation device. They also show good tolerance to both CO_2 and steam. The difficulty comes in the selection of an electronically conducting material. The cost of noble metals makes their incorporation into commercial devices unlikely. Therefore, the electron-conducting phase is limited to a

ceramic material such as the doped perovskites. Many perovskite materials react deleteriously with both YSZ and Bi_2O_3, as evidenced in a plethora of SOFC literature. Cerium gadolinium oxide appears to be less reactive with perovskites and has thus received more attention as the oxygen-ion conducting component of a dual-phase membrane.

The perovskite materials which are considered suitable for the electron-conducting phase are based around the generic composition (Ln,Ae)TmO_3, where Ln is a lanthanide, most commonly lanthanum, Ae is an alkaline earth, most commonly calcium, strontium or barium, and Tm is a transition metal, most commonly Ti, Cr, Mn, Fe, Co [4] (or the nontransition element, Ga [5]). An example of such a dual-phase mixture is $Ce_{0.8}Gd_{0.2}O_{1.9}$ mixed with $La_{0.6}Sr_{0.4}Co_{0.2}Fe_{0.8}O_{3-\delta}$ [6]. The ratio of the two phases is important in determining the overall properties of the material.

In general, the dual-phase approach has not been aggressively adopted because of the relatively low ambipolar conductivity so far achieved and the low oxygen flux which is obtained with these materials. While CGO- and Bi_2O_3-based materials can in themselves have a relatively high ionic conductivity of about 0.1–0.5 S cm^{-1} at 750–900 °C [7], when coupled with a significant volume fraction of electron-conducting material which does not contribute to the ionic conductivity, the overall ambipolar conductivity is substantially decreased. The oxygen flux through the membrane can be increased by decreasing the thickness, as discussed later. However, even with very thin membranes (<10 μm), the ionic conductivity needs to be greater than about 1 S cm^{-1} to approach desired commercial targets.

As a result of high oxygen-ion conductivity targets, the majority of attention has been directed toward the acceptor-doped perovskites and brownmillerites. These materials have oxygen diffusivities (which are directly related to the oxygen-ion conductivity) substantially greater than YSZ and CGO materials with the fluorite structure. They also have the added advantage in that they do not require a second phase to supply the necessary electron conductivity needed for charge balance of the oxygen anions.

As an example, the oxygen self-diffusivity of YSZ at 800 °C is approximately 5×10^{-8} cm^2 s^{-1} [8, 9] whereas $La_{0.6}Ca_{0.4}Co_{0.8}Fe_{0.2}O_{3-\delta}$ is near 1×10^{-7} cm^2 s^{-1} [10] and $Sm_{0.5}Sr_{0.5}CoO_{3-\delta}$ is more than 5×10^{-7} cm^2 s^{-1} at 800 °C [11].

When considering materials for oxygen separation and collection, the most important properties are: oxygen-ion conductivity, electron conductivity, creep resistance, thermal expansion, chemical stability, chemical expansion, oxygen surface exchange rate, thermal shock resistance, CO_2 tolerance, steam tolerance, sinterability, and absence of phase transitions within the operating temperature range (25–900 °C).

The properties of the (Ln,Ae)TmO_3 perovskite materials are strongly controlled by the dopants which are used. The choice of element for the A-site of the perovskite has some impact on the properties, but there is a more dramatic impact created by changing the ratio of lanthanide and alkaline earth.

The most common lanthanide selected is lanthanum. This can be readily substituted with samarium. The alkaline earths are typically selected from calcium,

strontium and barium. In the case of an iron-based perovskite with the formula $Ln_{0.6}Ae_{0.4}FeO_{3-\delta}$ the following matrix can be created:

$$\begin{array}{lll} La_{0.6}Sr_{0.4}FeO_{3-\delta} & La_{0.6}Ca_{0.4}FeO_{3-\delta} & La_{0.6}Ba_{0.4}FeO_{3-\delta} \\ Sm_{0.6}Sr_{0.4}FeO_{3-\delta} & Sm_{0.6}Ca_{0.4}FeO_{3-\delta} & Sm_{0.6}Ba_{0.4}FeO_{3-\delta} \end{array} \quad (6.1)$$

In every case, the material properties will be different, but not dramatically so. However, in the case of one elemental composition, varying the ratio of the components can have a much larger effect. In the case of $(La,Sr)FeO_{3-\delta}$, varying the La/Sr ratio from 10/1 to 1/10 will have a much larger effect. A group of materials with the formula $(La,Sr)FeO_{3-\delta}$ are listed below:

$$\begin{array}{lll} La_{0.9}Sr_{0.1}FeO_{3-\delta} & La_{0.8}Sr_{0.2}FeO_{3-\delta} & La_{0.6}Sr_{0.4}FeO_{3-\delta} \\ La_{0.4}Sr_{0.6}FeO_{3-\delta} & La_{0.2}Sr_{0.8}FeO_{3-\delta} & La_{0.1}Sr_{0.9}FeO_{3-\delta} \end{array} \quad (6.2)$$

Comparison of the impact of composition on some of the important membrane properties is listed in Table 6.2. (The number 5 represents the highest value and 1 represents the lowest.)

There is little difference in the properties if lanthanum is substituted with samarium. The perovskites which contain samarium tend to have a very slightly higher oxygen diffusivity and oxygen surface exchange rate relative to their lanthanum counterparts. However, due to the cost of samarium and the limited improvement obtained, most work is conducted on the lanthanum materials.

Tab. 6.2 Effect of A-site doping on perovskite materials properties, 1 is lowest, 5 is highest.

Composition	Oxygen-ion conductivity	Electronic conductivity	Thermal expansion	Oxygen surface exchange	CO_2 tolerance
$La_{0.6}Sr_{0.4}FeO_{3-\delta}$	3	3	3	3	3
$La_{0.6}Ca_{0.4}FeO_{3-\delta}$	3	3	3	3	3
$La_{0.6}Ba_{0.4}FeO_{3-\delta}$	4	3	3	3	2
$Sm_{0.6}Sr_{0.4}FeO_{3-\delta}$	3	3	3	3	3
$Sm_{0.6}Ca_{0.4}FeO_{3-\delta}$	3	3	3	3	3
$Sm_{0.6}Ba_{0.4}FeO_{3-\delta}$	4	3	3	3	2
$La_{0.9}Sr_{0.1}FeO_{3-\delta}$	1	1	2	1	3
$La_{0.8}Sr_{0.2}FeO_{3-\delta}$	1	1	2	2	3
$La_{0.4}Sr_{0.6}FeO_{3-\delta}$	2	2	3	3	3
$La_{0.2}Sr_{0.8}FeO_{3-\delta}$	4	4	3	3	2
$La_{0.1}Sr_{0.9}FeO_{3-\delta}$	4	4	3	4	2

Calcium, strontium and barium each have slightly different impacts on the properties of the perovskite. This is due, in part, to the difference in ionic radii of the ions. Calcium has a radius of 1.36 Å, whereas the radii of strontium and barium are 1.44 and 1.65, respectively. Thus barium-containing perovskites have a more open crystal lattice and thus have a higher ionic conductivity compared to strontium- and calcium-based materials. However, barium is more reactive towards CO_2-containing gas mixtures and forms barium carbonate very readily.

The substitution of the alkaline earth for lanthanum in different amounts has great impact on the electron-conducting properties. This is principally because strontium (and calcium or barium) has a valence of 2+ whereas lanthanum exists in the 3+ form. Thus, as strontium is substituted for lanthanum, there arises a need to equilibrate the charge of the cations and oxygen anions in the perovskite. This is done through a combination of creating oxygen vacancies in the lattice and changing the valence on the transition metal cation. As more Sr^{2+} replaces La^{3+}, the iron cation will change from Fe^{3+} to Fe^{4+} and more oxygen anion vacancies will form. The impact of increasing the level of strontium doping is to increase both the ionic conductivity due to more oxygen vacancies and the electronic conductivity as there are more Fe^{4+} ions. Both of these effects improve the oxygen transport properties of the membrane. Thus much work on the membrane materials centers around highly doped perovskite materials. However, there are some downsides to highly doping the perovskite. The highly doped materials contain many oxygen vacancies, thus reducing the creep strength of the material. Also, the number of Fe^{4+} cations (which will change their valence as the material is heated and loses oxygen) has an impact on the thermal and chemical expansion coefficients. This is shown below in Kröger–Vink notation:

$$2Fe_{Fe}^{x} + O_{O}^{X} \xrightarrow{\text{heat}} {}^{1}/_{2}O_{2(g)} + V_{O}^{\bullet\bullet} + 2Fe_{Fe} \tag{6.3}$$

Fe_{Fe}^{x} represents an Fe^{4+} cation in the doped perovskite, and $2Fe_{Fe}$ represents an Fe^{3+} ion occupying the Fe^{4+} site. The higher the level of strontium doping, the more Fe^{4+} cations will be present in the lattice. Thus, as the material is heated and loses oxygen from the lattice, Fe^{4+} cations are reduced to Fe^{3+}. This oxygen loss creates an expansion over and above that due to thermally induced vibrations. A similar effect is seen if the oxygen partial pressure is reduced. Thus, when a membrane material is exposed to a varying oxygen partial pressure on one side, a chemical expansion or contraction will occur if the transition metal ions change their valence. When the material has mainly Fe^{3+} cations, as in $La_{0.9}Sr_{0.1}FeO_{3-\delta}$, fewer iron ions change their valence state and thus the chemical expansion is reduced.

Another issue in choosing materials which have very high levels of strontium, calcium or barium doping is reaction with CO_2. While the absolute tolerance to CO_2 is heavily dependent on the transition metal selected, the higher the fraction of the carbonate-forming species present, the lower the tolerance to CO_2. In general, most iron-based perovskites have good CO_2 stability up to the limits of exposure in an oxygen separation device. It is also well established that barium is the

least tolerant to CO_2 of the alkaline earth dopants, whereas strontium is the most tolerant. There are complex equilibria which can be studied to show the effect of temperature and partial pressure of CO_2 on the alkaline earth oxides, and conditions can occur for which a strontium-containing perovskite has worse CO_2 tolerance than a barium perovskite. However, this is rare and, in general, does not occur under practical operating conditions.

Due to the positive and negative effects of the different doping cations, there is the option of using more than one dopant material. While there are relatively few academic publications on materials which contain (La,Sr,Ca,Ba)TmO_3, there has been significant patent activity including multiple-dopant substitution of the A-site [4, 5, 12, 13]. The reasons for mixing multiple dopants is to maximize the desired properties while attempting to reduce the undesired properties. For example, barium doping will have the effect of increasing the oxygen flux through the membrane, but, in turn, it will decrease the tolerance to CO_2. Therefore, mixtures of strontium and barium would be used as dopants.

There is an even greater impact on the materials properties governed by the selection of the transition metal cations. The selection of titanium in a material such as strontium titanate ($SrTiO_3$) results in a material which has negligible ionic conductivity, oxygen surface exchange, and electronic conductivity, but low thermal expansion, low chemical expansion and creep rate, whereas a cobalt perovskite such as $Sr_{0.7}La_{0.3}CoO_{3-\delta}$ [14] has very high oxygen diffusivity, oxygen surface exchange rate, and electronic conductivity, but a high thermal and chemical expansion and a high creep rate.

In general, the material properties of interest follow a systematic trend, shown in Table 6.3. The first element listed results in the highest value parameter, and the last is the lowest value.

There are several important factors in this table which influence the results. The first is that titanium is difficult to form into a perovskite which contains a significant amount of a lanthanide because titanium forms a 4+ cation which is

Tab. 6.3 Effect of B-site ion on materials properties, first element has highest property.

Property	Elements
Oxygen ion conductivity	Co, Fe, Ga, Mn, Cr, Ti
Electronic conductivity	Co, Fe, Mn, Cr, Ga, Ti
Oxygen surface exchange rate	Co, Fe, Ga, Mn, Cr, Ti
Thermal expansion coefficient	Co, Fe, Mn, Cr, Ga, Ti
Chemical expansion coefficient	Co, Fe, Mn, Cr, Ga, Ti
CO_2 tolerance	Ti, Ga, Cr, Mn, Fe, Co
Resistance to creep	Ti, Ga, Cr, Mn, Fe, Co

difficult to reduce and thus maintain charge neutrality. Another interesting feature is that gallium (which is not a transition metal and which does not possess multiple valence) falls outside the pattern of the other elements. A transition metal which produces a high ionic conductivity normally also produces a high electronic conductivity, due to multiple valence and a high oxygen surface exchange rate, but with poor CO_2 tolerance and a high creep rate. Gallium perovskites show high ionic conductivity but very low electronic conductivity, low thermal expansion rates and good CO_2 resistance. This makes perovskites with added gallium very interesting candidates for oxygen transport membranes [5, 15]. One downside to the use of gallium is the cost of the material.

A final point to note from this table is that the properties of the materials vary tremendously from one cation to another. A cobalt-based perovskite could have an oxygen diffusivity as much as 10 orders of magnitude higher than a chromium-based perovskite which contains the same A-site elements, but the thermal expansion coefficient could be more than double that of the chromate.

A leading choice of transition metal for the B-site of the perovskite is cobalt. Because the requirements for ionic conductivity are so stringent, it is often desired that the perovskite material selected contain cobalt on the B-site. The materials are also usually heavily doped to create the required oxygen vacancies in the high oxygen partial pressure environment of an air separation membrane. Thus, there has been considerable study of materials of the general formula $La_{1-x}Sr_xCoO_{3-\delta}$ such as $La_{0.3}Sr_{0.7}CoO_{3-\delta}$ [14, 16] and $Sr_{0.95}La_{0.05}CoO_{3-\delta}$ [17]. However, cobaltites certainly have their downsides. The resistance to reaction with CO_2 is very poor, the thermal expansion coefficient is very high, and the creep rate is very high. In addition, the extremely high dopant levels can lead to a situation in which the material transforms from a cubic structure at operating temperature to a hexagonal structure at room temperature. This phase transition has an associated volume change.

The combination of a high thermal expansion coefficient and a high creep rate is particularly difficult to overcome. Because the material will be subjected to large pressure gradients at high temperatures, creep resistance is essential. One method of increasing the overall resistance to creep is to place the dense membrane material onto a porous support of a different composition which has good creep resistance. However, most creep-resistant materials have a low thermal expansion coefficient (YSZ ~10.5 ppm K^{-1}, Al_2O_3 ~8 ppm K^{-1}) whereas $La_{0.3}Sr_{0.7}CoO_{3-\delta}$ has a thermal expansion coefficient of more than 20 ppm K^{-1}.

The result is that a tremendous amount of work has occurred in which two and sometimes three transition metal cations are selected for the B-site elements [12, 13]. The most commonly studied have used the addition of iron. This LSCF (lanthanum, strontium, cobalt, iron) perovskite also finds use as a solid oxide fuel cell cathode.

Iron is an obvious choice because it ranks high in terms of ionic and electronic conductivity, while having a lower thermal expansion coefficient, higher creep resistance and higher tolerance towards CO_2. Materials such as $La_{0.2}Sr_{0.8}Co_{0.8}Fe_{0.2}O_{3-\delta}$ are regularly described, and their performance as membranes measured.

The absolute oxygen fluxes through these materials may not be as high as in a cobaltite, but they have better robustness for the oxygen separation process.

Other additions which work well are gallium and also titanium. The addition of manganese to a cobaltite decreases the performance dramatically, whereas adding chromium increases the difficulty in fabrication.

Other work which has occurred in the development of perovskite materials for oxygen separation has been on making the materials A-site deficient. In other words, creating a perovskite which instead of having the formula ABO_3 has the formula $A_{1-x}BO_3$ [4, 18, 19]. Some of this work stems from the fact that if the materials have excess A-site cations, then any lanthanum present becomes extremely hygroscopic. Due to the complexity of fabricating these multicomponent metal oxides, it is, therefore, often desirable to deliberately reduce the A-site cations by 1–2% to minimize the chance of making a material with A-site excess. However, there has also been a line of study which shows that by making an A-site deficient perovskite, the material becomes more tolerant to CO_2 as well as to moisture [19]. This is particularly beneficial in cases for which the membrane will be operated under a direct-fired condition and for which the partial pressure of CO_2 can be substantial.

6.2.3 Membrane Architecture Concepts

Membrane architecture concepts are as varied as the material concepts and are equally important in determining the success of an oxygen transport device. The architecture can be considered from a membrane standpoint (thick dense membrane, supported thin film, sandwich structure) and from a system standpoint (tubular, planar, monolith).

It is very clear that due to the high desired oxygen transport performance requirements of the membrane, a supported thin film is necessary to reach the target oxygen flux. Early work was conducted by Teraoka in which he compared the performance of a dense thick perovskite to that of a thin film perovskite of the same composition supported on a porous support [20]. The concept is similar to that utilized in solid oxide fuel cells in which the design is often to use an anode-supported electrolyte. However, the benefits to an oxygen separation device of using a supported thin film are even greater. In the case of a solid oxide fuel cell, a porous anode is required to allow the binary diffusion of fuel gases to the electrolyte and combustion product gases away from the electrolyte. This binary diffusion is relatively slow, and so-called concentration polarization resistances are common in solid oxide fuel cells. However, in an oxygen separation device, there is no binary diffusion in the permeate but only the viscous flow of oxygen away from the membrane. Thus, with a well-designed pore structure in the membrane, gas phase mass transfer through the porous support is rarely an issue. However, it is important to maximize the surface area of the porous support near the interface with the dense separation layer [12]. Therefore, there are efforts to create graded pore structures in which a fine porous structure occurs near the dense

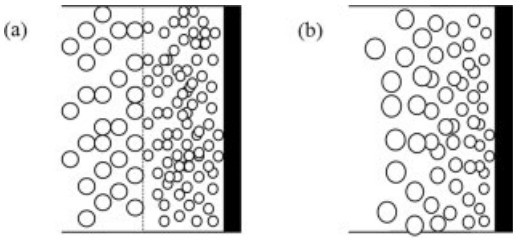


Figure 6.3 (a) Step change in pore size of support structure. (b) Continuous change in pore size of support structure.

membrane interface that transitions either stepwise or gradually to a larger pore size away from the dense membrane. Two examples are shown in Fig. 6.3 (a) and (b).

The step change in pore structure can be fabricated in discrete steps. It also allows the use of dissimilar materials for the porous structure, as used in early Westinghouse SOFC cathode designs and discussed by Thorogood et al. for oxygen transport membranes [12].

In addition to a suitable porous structure on the permeate side of the membrane to facilitate oxygen transfer out of the membrane, additional enhancements to the surface of the feed side of the membrane may be necessary. The oxygen surface exchange coefficient of cobalt perovskite materials results in a limitation to the oxygen flux once the membrane is below a certain thickness. The membrane critical thickness is defined as the point at which the resistance due to oxygen surface exchange (e.g., adsorption and dissociation of molecular oxygen) is equal to the resistance due to bulk oxygen diffusion. The critical thickness is defined as $T_c = D^*/k$, where D^* is the oxygen self-diffusion coefficient, and k is the oxygen surface exchange coefficient. In general, this critical thickness is between 10 and 100 μm. As an example, the material, $Sm_{0.5}Sr_{0.5}CoO_{3-\delta}$, has $D^* = 1.5 \times 10^{-7}$ cm^2 s^{-1} at 700 °C and $k = 1.5 \times 10^{-5}$ cm s^{-1} at 900 °C, yielding a T_c of 100 μm. Therefore, when membrane thicknesses are decreased to improve oxygen flux, improvements to the effective oxygen surface exchange rate are required. The simplest method is to provide an extended surface area on the membrane by creating a porous structure. In effect, a sandwich is created with the dense membrane in the middle and two porous materials on either side, which is similar to that of a solid oxide fuel cell.

In a solid oxide fuel cell, there are cases for which the cathode is thick and the anode thin (e.g. Siemens-Westinghouse) and cases for which the anode is thick and the cathode is thin (e.g. General Electric). For a solid oxide fuel cell, there is no significant gain to be made in terms of the oxygen transport by switching from anode-supported to cathode-supported. Much depends on the operation parameters of the fuel cell. In the case of an oxygen separation device, there is a very large disparity in the oxygen transport properties, depending on the side on which the porous support is placed. On the permeate side, only one gaseous species is present – oxygen flowing away from the dense membrane. On the feed side, oxygen

needs to diffuse through the porous structure to the dense separation membrane, and nitrogen (and argon) needs to diffuse away from the dense membrane to the feed exhaust stream. Diffusion in this circumstance is much slower than the simple viscous flow of oxygen away from the membrane. In addition, the oxygen flux in an oxygen transport membrane is much higher than in a solid oxide fuel cell (approximately 5 or 10 times higher). As a result, air-side mass transfer limitations become increasingly important to the oxygen flux. Therefore, the porous layer used to enhance the oxygen surface exchange rate on the air side should be kept thin, preferably in the 20–50 μm range.

A preferred embodiment of an oxygen transport membrane would thus have a thin porous support on the feed side to improve oxygen exchange, a thin dense separation membrane, a fine pore structure interfacial layer to facilitate oxygen transfer out of the membrane and a coarse porous support to maximize product flow and provide the structural support. An example is shown in Fig. 6.4. The coarse porous support material could be made out of inert material because it is not chemically active in the transport of oxygen. This allows the use of less expensive materials which may also have better strength characteristics.

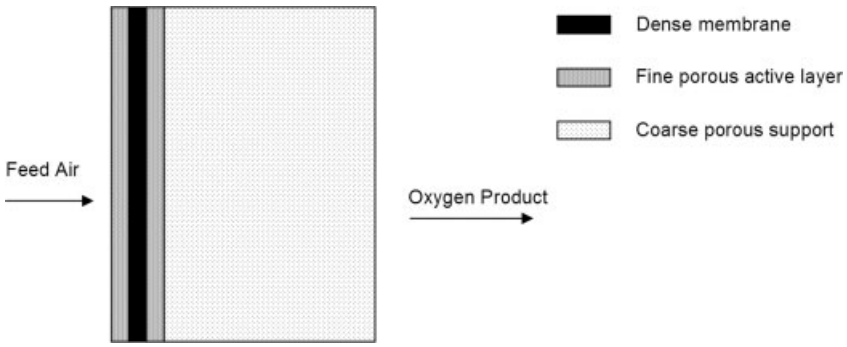


Figure 6.4 Oxygen transport membrane structure.

In the case of a solid oxide fuel cell, the anode or cathode support can be relatively thin because the component does not need to bear a very high load. However, in the case of oxygen transport membranes, the porous support needs to withstand a differential pressure of 20 atm or greater. Therefore, porous supports which are several millimeters thick are often considered. Alternatively, other concepts for strengthening the support structure are considered. These can include internal structures such as multichannel tubes, distinct solid porous inserts in tubes [21] and support braces in planar geometries. Examples of such structures are shown in Fig. 6.5 and 6.6.

Figure 6.5(a) shows a membrane supported by three distinct porous layers. The central porous support uses a solid, porous body with multiple channels. Figure 6.5(b) shows a simpler device with only two distinct porous supports. The central support is a solid porous body. This is only suitable if the gas diffusion

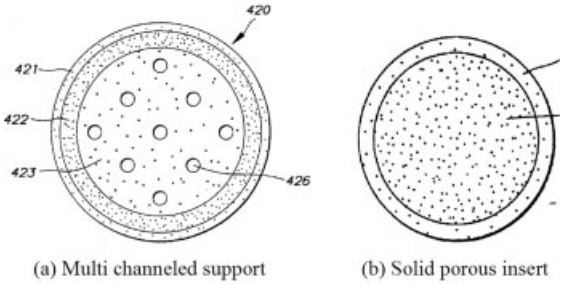


Figure 6.5 Internal support structures in tubular geometry.

rate is very high such that diffusion down the length of the tube does not substantially reduce the oxygen flux through the device.

The upper diagram in Fig. 6.6 shows a simple device with a membrane on a porous support with a solid, porous body filling the sandwich. The central diagram is similar in concept except that there are distinct multiple layers of porous bodies. In both cases, however, there are no open channels, and the gas must diffuse along the length of the solid porous support to an exit channel. The lower diagram shows a membrane with multiple porous supports with the center piece containing channels to enable gas to diffuse out of the device to a collection manifold. This decreases the resistance to gas phase mass transfer of the device.

Internal structures within the membrane architecture add extra complexity to the fabrication process and operation of the membrane. Therefore, careful consideration needs to be given to determination of the best approach to maximize the

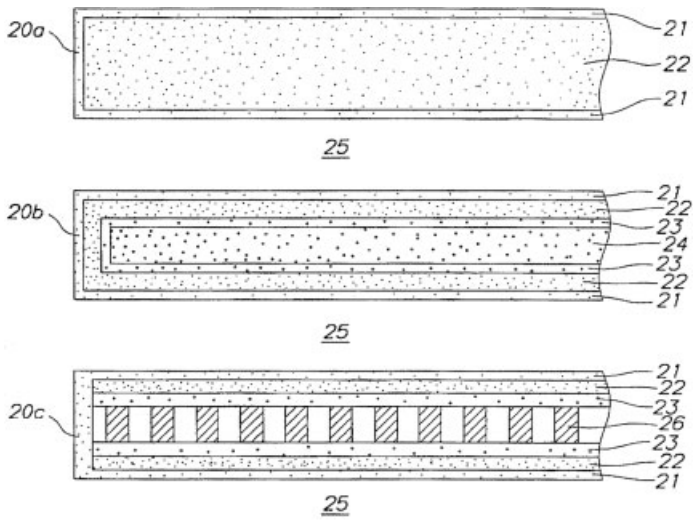


Figure 6.6 Internal support structures in planar geometry.

life of the membrane under the large pressure gradient which occurs during operation.

The next level up from the basic membrane architecture is the selection of membrane shape. The main two architectures studied have used tubular geometries and a modified planar geometry. Both have their advantages and disadvantages for oxygen separation.

A tubular geometry is a relatively simple geometry to fabricate and operate. Porous support tubes can be produced on a large scale through extrusion or isostatic pressing at relatively low cost. The dense membrane can be applied using conventional technology such as dip coating. Other technologies have been developed such as multiple layer isostatic pressing [24], using a tape cast layer wrapped around the tubular support, followed by pressing and sintering [25].

Sealing of the materials is difficult due to the high pressure differential, but the sealing area per unit area of membrane surface is small in a tubular design. The stresses which are generated by the pressure gradient in a tubular design are relatively symmetrical and result in the ceramic being under a state of compression. However, creep buckling of tubes can occur if the support is not sufficiently robust to tolerate the combination of high temperature and pressure.

One downside of the tubular design is that it results in an expensive system. The membrane surface area per unit volume is low compared to a planar structure. In addition, every tube needs to be sealed to the metal manifold, and this is expensive. To reduce the cost, the simplest way is to increase the length of the membrane tubes. Unfortunately, manufacturing yield suffers as tube length goes up, and the reliability of the ceramic tubes during operation goes down. To maximize the surface area per unit volume, the tube diameter should be reduced. However, long tubes which have a small diameter are subject to Euler buckling when exposed to the high pressure on the air side of the oxygen separation device. Larger diameter tubes suffer from a higher hoop stress, creating additional problems with long term reliability. It is apparent that there are many conflicting parameters when selecting the architecture for a tubular geometry, and an optimum configuration needs to be determined depending on the material properties of the membrane itself. Unfortunately, the general trend is that methods to improve robustness and reliability result in an increase in the cost of the device. Table 6.4

Tab. 6.4 Effect of tubular dimensions on component costs and durability.

	Manufacturing cost (cost/area)	Installation and operating cost	Compressive creep resistance	Euler buckling resistance
Increase length	decrease	decrease	no change	decrease
Increase diameter	decrease	increase	decrease	increase
Increase wall thickness	increase	no change	increase	increase

shows the impact of changing design parameters on the cost and robustness of the device. The manufacturing cost does not include any change in yield which may occur as a result of making changes to the architecture.

Study of the table can lead to several conclusions as to the preferred architecture. If material robustness can be improved and manufacturing yield kept high, then long, thin tubes with a small diameter will lead to the most cost-effective approach. However, given the constraints of very high oxygen-ion conductivity, the materials are generally not very strong, and so the length of the tube must be kept relatively short.

A conventional planar geometry, which has been widely adopted in solid oxide fuel cell technology, has very significant problems in the case of a pressure-driven oxygen separation membrane. The main reason is the large pressure differential. The large seal areas which are present in planar designs are very difficult to effectively seal against a 20 bar pressure gradient. There are also large bending moments placed on the planar design of the oxygen transport membrane leading to early failures.

There have been very clever developments by Air Products and Chemicals to adapt the planar design for oxygen separation membranes. The reason for the effort is that the cost of manufacturing and operating a planar design has the potential to substantially reduce the cost of the device. A much larger membrane surface area per unit volume can be achieved in a planar design than in a tubular design. This adapted planar device is shown in Fig. 6.7. It has the benefit of a large surface area to volume ratio, but reduces the large sealing areas and bending stresses which are typically associated with planar designs.

The system has the benefits of low-cost manufacturing potential as well as a high surface area to volume ratio, and low sealing area. It also avoids some of the buckling issues associated with tubes.

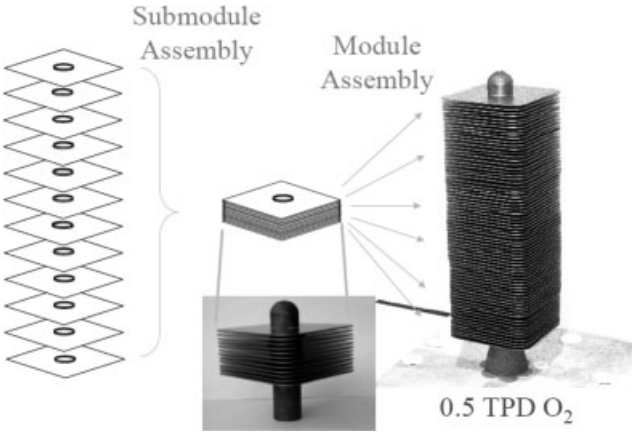


Figure 6.7 Stacked planar geometry.

6.2.4
Summary of Oxygen Separation Materials and Architecture

In summary, there has been tremendous progress in the development of both materials and architectural designs for oxygen separation. The challenges which are associated with commercial oxygen production are still present, and further development of materials in conjunction with membrane design are required to enable commercial success.

6.3
Syngas Production and Combustion Applications

We will now consider the development activity in the application of syngas production and combustion applications. As in the case of oxygen separation and collection, a range of materials has been examined in this area including perovskites, brownmillerites, fluorites and composites of various structures. Given the similarity of high temperatures and high pressure gradients which are present as in oxygen separation, architecture developments are similar to that used in oxygen separation.

6.3.1
Background for Selection of Materials for Syngas Production and Combustion Applications

For syngas production, the temperature requirements are high (>1000 °C), pressure differentials are high, 300–450 psi (20.7–31.0 bar), highly reducing gases are present on one side of the membrane, and oxidizing gases are present on the other side. High steam and CO_2 concentrations are also present. Many of these conditions are similar to those experienced by interconnect materials in solid oxide fuel cells. The exception is the pressure differential. However, unlike an interconnect material which requires a high electronic conductivity and a low oxygen-ion conductivity, materials for syngas production require both high electron and oxygen-ion conductivities. A high oxygen surface exchange rate on the air side and a high fuel oxidation rate on the fuel side are required. An advantage which syngas production has over oxygen separation is that the driving force for oxygen diffusion is orders of magnitude greater. In oxygen separation, an average oxygen partial pressure gradient is approximately 3, whereas in syngas production, it is about 10^{16}. This is due to the extremely low oxygen partial pressure present in the syngas.

Combustion applications, such as in boilers or process heaters, have somewhat different requirements relative to syngas production. The first is that the fuel utilized needs to be fully oxidized to CO_2 and H_2O rather than partially oxidized to CO and H_2. This implies that at the exhaust end of the reactor there is no longer a very large oxygen partial pressure gradient. Typical oxygen partial pressure of a

Tab. 6.5 Desired materials properties for ceramic oxygen transport membranes used for syngas production.

Property	Maximum	Minimum	Preferred range
Oxygen ion conductivity (S cm^{-1})N/A	0.2	0.5–2	
Electronic conductivity (S cm^{-1})	N/A	5	20–500
Oxygen surface exchange coefficient (cm s^{-1})	N/A	10^{-8}	10^{-6}–10^{-8}
Thermal expansion coefficient (ppm K^{-1})	20	8	12–17
Chemical expansion coefficient (%)	0.25	–0.25	0–0.1
Creep rate (cm s^{-1})	10^{-8}	N/A	10^{-9}–10^{-11}
CO_2 tolerance (bar CO_2)	N/A	0.5	0.6–2.0

Tab. 6.6 Desired materials properties for ceramic oxygen transport membranes used for combustion applications.

Property	Maximum	Minimum	Preferred range
Oxygen ion conductivity (S cm^{-1})N/A	0.2	0.5–2	
Electronic conductivity (S cm^{-1})	N/A	5	20–500
Oxygen surface exchange coefficient (cm s^{-1})	N/A	10^{-8}	10^{-6}–10^{-8}
Thermal expansion coefficient (ppm K^{-1})	20	8	12–17
Chemical expansion coefficient (%) 0.2	–0.25	0–0.1	
Creep rate (cm s^{-1})	10^{-7}	N/A	10^{-8}–10^{-10}
CO_2 tolerance (bar CO_2)	N/A	0.3	0.4–0.7

CO_2 and H_2O gas mixture present in an exhaust stream is about 0.02 bar, due to the presence of about 2% oxygen. The excess oxygen is present to ensure that there are no uncombusted hydrocarbons or CO. Another major difference is that a high fuel pressure is not required because the products of the combustion are not collected for further use. Tables 6.5 and 6.6 summarize the materials proper-

ties which are required for successful operation of a syngas production ceramic oxygen membrane and a combustion application membrane.

It should be noted that the chemical expansion is absolute. Therefore, the desired chemical expansion per unit of partial pressure needs to be much less than in oxygen separation.

6.3.2 Membrane Materials Concepts

The basic materials concepts are very similar to those discussed for oxygen separation. Fundamentally, the two types of device are the same. The first variation is the dual-phase material which contains one component that is an ionic conductor and one that is an electronic conductor. The second variation is the single-phase material which is both an electronic and ionic conductor.

Due to the low oxygen partial pressure to which the membrane is exposed on the fuel side, selection of single-phase perovskite-type materials is limited. As discussed earlier, cations used at the B-sites of a perovskite include cobalt, iron, gallium, manganese, chromium and titanium. Perovskites which contain cobalt will usually be reduced in a fuel environment, creating substantial chemical expansion or even structural breakdown. Iron also tends to be reduced in low oxygen partial pressure environments, but less so than cobalt. High chemical expansion still exists with iron-based perovskites, but structural breakdown is less likely. Gallium-, chromium- and titanium-containing perovskites are relatively stable in reducing atmospheres and do not undergo substantial chemical expansion. Manganese perovskites can also be reduced and create problems in syngas applications.

Of gallium, chromium and titanium B-site cations, only gallium perovskites have substantial oxygen-ion conductivity. Therefore, if chromium or titanium are used as part of a single-phase mixed electronic-ionic conducting membrane, they need to be used in conjunction with iron or small amounts of cobalt [26]. Gallium perovskites, while possessing potential for excellent ionic conductivity, do not have sufficient electronic conductivity and need the addition of iron, cobalt or chromium.

As in the case with oxygen separation materials, lanthanum is the preferred lanthanide. Strontium, calcium and barium are all possible dopants, although the susceptibility of barium to reaction with CO_2 limits its use. Efforts to make the materials A-site deficient are also common.

An example of a complex material used in syngas applications is $La_{0.2}Sr_{0.8}Fe_{0.8}Co_{0.1}Cr_{0.2}O_{3-\delta}$ [26]. This composition contains many of the features previously discussed. It is predominately an iron-based perovskite, providing both electronic and ionic conductivity. In order to improve the chemical stability and resistance to chemical expansion, the B-site is doped with chromium. A small amount of cobalt is added in order to improve the electronic conductivity, and cobalt will also increase the ionic conductivity slightly. Strontium is used as the A-site dopant in order to avoid the problems associated with calcium, and particularly with barium, in regard to reaction with CO_2. In addition, the material has been made substantially A-site deficient to improve its stability.

These relatively complex perovskites, while solving many problems, suffer from others. The complex nature of the material makes manufacturing on a large scale somewhat expensive. The material is also on the threshold of instability in the high temperature and low oxygen partial pressure environment to which it is exposed. There is also some concern about the creep rate of similar perovskite materials.

An alternative approach is to use two-phase composite membranes. In this case, a very stable ionic conductor can be used, such as YSZ, CGO or lanthanum gallate perovskites. The ionic conductor is then mixed with an electronic conductor such as a noble metal or an electron-conducting perovskite such as lanthanum chromite. This approach is better suited to the fuel-gas type applications for which the requirements of oxygen-ion conductivity are reduced compared to the oxygen separation application. However, the temperatures of operation are higher in combustion and syngas applications, thus increasing the concern for possible reaction between the two phases of the materials. In both the single-phase and dual-phase applications, there is still a need for progress in materials development to help facilitate commercial operation of either a syngas production facility or a combustion device.

6.3.3 Membrane Architecture Concepts

Many of the options for different membrane and system architectures have been covered in the section on oxygen separation materials. The advantages and disadvantages of planar and tubular designs are very similar whether the device is to be used for making syngas or separating oxygen. One difference is that in the case of separating oxygen, only one feed gas is required (air). In the case of syngas or combustion, two feed gases are required (air on one side and fuel on the other). This increases the complexity of the manifolding and should be considered when selecting a membrane architecture.

As with solid oxide fuel cells, a thin membrane is preferred to reduce the resistance to oxygen transfer. Therefore a support is required. Unlike the case of oxygen separation, for which it is more straightforward to design the support system, when two feed gases are required, the relative gas diffusion rates must be considered in determining the side on which to place the support. For a syngas application, for which the fuel feed gas is natural gas and the products are H_2 and CO, the thick porous support should be placed on the fuel side. Simplistically, this is because hydrogen can diffuse through the porous support substantially faster than the oxygen can diffuse through an air-side porous support. In the case of combustion applications, in which the fuel is completely oxidized to CO_2 and H_2O, and for which the feed fuel gas is substantially depleted, the gas diffusion rates are slower for the fuel and combustion products. This is more akin to the case in a solid oxide fuel cell. For combustion, creation of the best support architecture, which will be dependent on actual operating conditions, may not be so straightforward.

6.3.4 Summary of Syngas and Combustion Applications Materials and Architecture

Materials development for syngas and combustion is ongoing. There has been a tremendous effort in the development of both single-phase materials with multiple co-doping and different manufacturing methods, as well as efforts in the development of dual and even three-phase mixtures to overcome the substantial difficulties associated with operating ceramic membranes in the harsh conditions of a syngas device. Similarly, architecture developments continue to improve the robustness of the devices as well as to lower the cost of production. Work in both areas continues, and more developments will be made.

References

1 T. J. Mazanec, T. L Cable, J. G. Frye, Electrocatalytic cells for chemical reaction, *Solid State Ionics* **1992**, *53–56*, 111.
2 T. J. Mazanec, T. L Cable, J. G. Frye, Jr., U.S. Pat. 4 793 904, Dec. 27, 1988.
3 M. Liu, A. V. Joshi, Y. Shen, K. Krist, A. Virkar, U.S. Pat. 5 478 444, Dec. 26, 1995.
4 T. J. Mazanec, T. L Cable, J. G. Frye, Jr., W. R. Kliewer, U.S. Pat. 5 306 411, Apr. 26, 1994.
5 R. Mackay, M. Schwartz, A. F. Sammells, U.S. Pat. 6 592 782, July 15, 2003.
6 J. D. Sirman, J. C. Chen, U.S. Pat. 6 514 314, Feb. 4, 2003.
7 S. Wang, T. Kobayashi, M. Dokiya, T. Hashimoto, Electrical and Ionic Conductivity of Gd-Doped Ceria, SOFC VI, *Proc.-Electrochem. Soc.* **1999**, *99-19*, 193.
8 P. S. Manning, J. D. Sirman, R. A. DeSouza, J. A. Kilner, *Solid State Ionics* **1997**, *100*, 1.
9 H. Solomon, Ph.D. Dissertation, Université Paris, 1992.
10 S. Carter, A. Selcuk, R. J. Chater, J. Kajda, J. A. Kilner, B. C. H. Steele, *Solid State Ionics* **1992**, *53–56*, 597.
11 I. C. Fullarton, J. A. Kilner, B. C. H. Steele, P. H. Middleton, *Proceedings of the Second International Symposium on Ionic and Mixed Conducting Ceramics*, Electrochemical Society PV 94-12, p. 9, 1994.
12 R. M. Thorogood, R. Srinivasan, T. F. Yee, M. P. Drake, U.S. Pat. 5 240 480, Aug. 31, 1993.
13 T. J. Mazanec, T. L Cable, U.S. Pat. 6 488 739, Dec. 3, 2002.
14 M. van der Haar, Mixed-Conducting Perovskite Membranes for Oxygen Separation, Ph.D. Dissertation, Universiteit Twente, The Netherlands, 2001.
15 R. Mackay, M. Schwartz, A. F. Sammells, U.S. Pat. 6 165 431, Dec. 26, 2000.
16 R. H. E. van Doorn, H. Kruidhof, H. J. M. Bouwmeester, A. J. Burggraaf, *Mater. Res. Soc. Symp. Proc.* **1995**, *369*, 373–383.
17 T. J. Mazanec, T. L Cable, U.S. Pat. 5 648 304, July 15, 1997.
18 N. Q. Minh, *J. Am. Ceram. Soc.* **1993**, *76*, 563.
19 M. F. Carolan, S. A. Motika, P. N. Dyer, P. B. Alba, Eur. Pat. EP 0 732 306 B2, Mar. 13, 1996.
20 Y. Teraoka *J. Ceram. Soc. Japan. Int. Ed.* **1989**, *97*, 458.
21 B. A. van Hassel, R. Prasad, J. Chen, J. Lane, U.S. Pat. 6 565 632, May 20, 2003.
22 P. N. Dyer, J. D. Bright, M. F. Carolan, E. Minford, R. E. Richards, S. L. Russek, D. M. Taylor, M. A. Wilson, U.S. Pat. 5 599 383, Feb. 4, 1997.
23 D. M. Taylor, J. D. Bright, M. F. Carolan, R. A. Cutler, P. N. Dyer, E. Minford, D. W. Prouse, R. E. Richards, S. L. Russek, M. A. Wilson, U.S. Pat. 5 681 373, Oct. 28, 1997.
24 P. S. Apte, S. W. Callahan, J. S. Schneider, J. C. Chen, U.S. Pat. 6 776 941, Aug. 17, 2004.
25 J. C. Chen, V. M. Stawisuck, R. Prasad, U.S. Pat. 6 524 421, Feb. 25, 2003.
26 W. T. Stephens, T. J. Mazanec, H. U. Anderson, *Solid State Ionics* **2000**, *129*, 271.